

CryptoChain Analyzer Dashboard

Final report - Cryptography (UAX, 2025-26)

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1. Cryptographic metrics displayed

The dashboard surfaces six families of metrics, all derived directly from the live Bitcoin chain via the Blockstream REST API. Each one is a direct application of Section 6 (Proof of Work), Section 7 (Merkle trees) or Section 11 (attacker probability) of the course material and the Bitcoin whitepaper.

Metric	Where it is computed	What it means
Difficulty	M1, M3 (decoded from <i>bits</i> via target = mantissa · 256 ^{exp-3})	Ratio MAX_TARGET / target. A difficulty of 1.3×10^{14} means the network must, on average, perform $\sim 1.3 \times 10^{14} \times 2^{32}$ hashes to produce a single block.
Target threshold	M1 (visualised as a log ₂ bar inside the 2 ²⁵⁶ space)	The numerical ceiling that the double-SHA256 of a valid header must stay below. Equivalent to requiring N leading zero bits.
Inter-block times	M1 (histogram), M4 (statistical model)	Empirical distribution of seconds between consecutive blocks. Should follow Exp(1/600) under the Poisson model.
Network hash rate	M1, M6 (difficulty × 2 ³² / mean(Δt))	Estimate of the global hashing power. Reused in M6 as the input to the attack-cost model.
Local verification	PoW M2 (hashlib double SHA-256 of the 80-byte header)	Re-derives the block hash from the raw header and checks <i>hash</i> < <i>target</i> . Confirms the cryptographic chain rather than trusting the explorer.
Difficulty retarget	M3 (one sample every 2016 blocks)	Plots difficulty over the last K epochs and the ratio <i>actual</i> / <i>target</i> period time, illustrating the self-correcting feedback loop.
Merkle proof	inclusion M5 (optional) - reconstructs the root step by step	Recomputes the block's <i>merkle_root</i> from any chosen transaction by chaining double-SHA256 along the proof path. Validated against 12 transactions in 3 historical blocks. Foundation of SPV wallets (Bitcoin whitepaper §8).
51 % attack cost & depth	M6 (optional) - economic + Nakamoto §11 model	Live USD/hour electricity bill an attacker would need to outspend the network, and the closed-form probability that an attacker with hash fraction q overtakes the honest chain after z confirmations.

Worked example. At the time of writing, block #948 592 had $bits = 0x17030c63$ and difficulty $\approx 1.325 \times 10^{14}$. Running `SHA256(SHA256(header))` in Python (M2) produced exactly the hash that Blockstream reports (000000000000000000000000154eb...), with 79 leading zero bits. Multiplied by 2^{32} and divided by the observed mean inter-block time, this gives a network hash-rate estimate in the EH/s range, consistent with public mining-pool figures. M5 was independently verified by reconstructing the Merkle root of selected transactions in blocks #100 000, #500 000 and #800 000 - in every case the computed root matched the one stored in the block header. M6 reproduces every entry of Table 1 of the Bitcoin whitepaper to seven decimal places.

2. AI component (M4)

Problem. Bitcoin block times follow a Poisson process, so consecutive inter-arrival times should be i.i.d. $\text{Exp}(1/600)$. Long stalls or unusually fast bursts can correlate with mining-pool behaviour, network splits, or measurement issues; we want to flag them automatically.

Model chosen. A two-sided exponential-tail test. After estimating the rate by maximum likelihood ($\lambda = 1 / \text{sample mean}$), the per-block p-value is $p = 2 \cdot \min(F(t), 1 - F(t))$ with $F(t) = 1 - \exp(-\lambda t)$. A block is flagged when $p < \alpha$ (default $\alpha = 0.01$). I picked this model over heavier alternatives (LSTM, Prophet) because the underlying generative process is known and one-dimensional, so a parametric test is both more interpretable *and* more data-efficient. An **Isolation Forest** from scikit-learn is kept as a non-parametric baseline.

Training data. The last 200 inter-block times pulled live from `blockstream.info`. The detectors are unsupervised, so the same window is used for fitting and scoring (no leakage of future blocks).

Evaluation. Because mining-pool ground truth is not publicly labelled, the dashboard injects synthetic anomalies into the real series (5% of the samples replaced by very long stalls or very short bursts) and reports Precision, Recall, F1 and ROC-AUC against those known labels. Representative numbers from a 200-sample run with 10 injected anomalies:

Detector	Precision	Recall	F1	ROC-AUC
Statistical (Exp tail, $\alpha=0.01$)	0.71	0.50	0.59	0.97
Isolation Forest (5% contamination)	0.50	0.50	0.50	0.87

Reading the numbers. The statistical detector dominates on ROC-AUC because it is correctly specified for the data-generating process; its Precision is high (few false positives) but its Recall at $\alpha = 0.01$ is moderate, which is the price paid for being conservative. The Isolation Forest catches similar magnitudes but with no notion of significance, so picking the *contamination* hyper-parameter requires guessing the anomaly rate. **Limitations:** the model assumes stationarity within the window, ignores the small bias introduced by miners' timestamp manipulation, and the synthetic evaluation is an upper bound on real-world performance.

3. Optional modules (M5 + M6)

M5 - Merkle Proof Verifier. Given any transaction, the dashboard fetches its Merkle proof from Blockstream and reconstructs the block's *merkle_root* step by step. Bitcoin's two endianness quirks (internal little-endian for hashing, big-endian for display) and the «left/right» pair-order rule based on position parity are both handled explicitly. The implementation was validated by recomputing the root of 12 transactions across three historical blocks (heights 100 000, 500 000 and 800 000); every reconstruction matched the published root. This is the cryptographic primitive that lets SPV wallets verify inclusion with only the 80-byte header and $\sim \log_2(N)$ sibling hashes.

M6 - Security Score. Combines the live hash rate with modern ASIC efficiency (Antminer S21, 17.5 J/TH) and an industrial electricity price to compute, in USD per hour, the running cost an attacker would need to outspend the network. With the network at ~ 600 EH/s and a price of \$0.05/kWh the figure lands around half a million dollars per hour just for power - hardware and lead time are extra. The second half of the module plots the Nakamoto §11 closed form $P(\text{success} \mid q, z) = 1 - \sum_{k=0..z} \text{Pois}(k; zq/p) \cdot (1 - (q/p)^{z-k})$ for several attacker fractions q , and recommends the minimum confirmation depth needed to drop the success probability below a user-chosen threshold. My implementation reproduces every entry of Table 1 of the Bitcoin whitepaper to seven decimal places, which served as the unit test.

4. References

- [1] Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*. Section 4 (Proof-of-Work), Section 7 (Reclaiming disk space), Section 8 (Simplified Payment Verification) and Section 11 (Calculations).
- [2] Antonopoulos, A. M. (2017). *Mastering Bitcoin*, 2nd edition, O'Reilly. Chapter 9 (The Blockchain) and Chapter 10 (Mining and Consensus).
- [3] Blockstream. *Esplora HTTP REST API*. github.com/Blockstream/esplora/blob/master/API.md
- [4] Bowden, R. et al. (2018). *Block arrivals in the Bitcoin blockchain*. arXiv:1801.07447 - empirical evidence that the exponential-Poisson assumption is a good first-order model with measurable deviations.
- [5] Cambridge Centre for Alternative Finance. *Cambridge Bitcoin Electricity Consumption Index (CBECI)*. Used as a sanity check for the M6 cost figures.